Atharv Krishnat Patil

Junior Undergraduate | Department of Mechanical Engineering

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2024 - Present	BTech Mechanical Engineering	Indian Institute of Technology, Kanpur	8.9 /10.0
2024	HSC (XII)	Shraddha Jr. College, Ichalkaranji	93.67%
2022	SSC (X)	Jantara Kalpvruksha Vidya Mandir, Jaysingpur	97.80%

SCHOLASTIC ACHIEVEMENTS

- Honored with the **Academic Excellence Award** at **IIT Kanpur** in recognition of exemplary academic achievement *(2025)*
- Achieved a top-tier **SPI of 9.3** during **Semester III**, reflecting consistent academic proficiency and scholastic excellence *(2025)*

KEY PROJECTS

ARVSAL: Personal AI Operating System and Digital Companion *(Self Project)* — *Jan'26 - Present*

Objective	• To design **local-first Personal AI Operating System** capable of **long-term memory** and **workflow automation**
Approach	• Architected **modular multi-process system** with memory management, voice interface, automation workflows & local AI • Developed **deterministic intent routing** and command execution pipelines to reduce dependence on generative models • Built **voice-driven pipelines** incorporating speech recognition, activity detection, and conversational response generation
Result	• Unified **memory, voice, automation, and local AI capabilities** into a single operating-system-style platform • Continuous user assistance and personalization with **production-scale codebase exceeding 21,000 lines**

Arvsal-Gard: Real-Time File Protection & Malware Analysis System *(Self Project)* — *Mar'26*

Objective	• To design a lightweight **desktop security application** capable of background execution and malicious file detection
Approach	• Built a **real-time file monitoring pipeline** capable of scanning newly created and modified files for security threats • Implemented **signature-based detection** using cryptographic hashing and YARA-inspired rule matching techniques • Engineered **parallel scanning workflows** utilizing worker threads to improve responsiveness and reduce scan latency
Result	• Delivered a deployable **Electron-based desktop application** featuring real-time monitoring and quarantine management

AVA-Listener: Custom Wake-Word Detection SDK *(Self Project)* — *May'26 - June'26*

Objective	• To develop an **open-source wake-word detection SDK** for custom voice activation for ARVSAL and other AI assistants
Approach	• Architected a **cross-language inference** connecting Node.js with Python-based speech processes through **WebSocket** • Integrated **ONNX-based speech recognition models** and implemented custom wake-word matching strategies • Designed an installable **NPM package** with automated model acquisition, caching, and verification mechanisms
Result	• Released AVA-Listener as a publicly available **NPM package** supporting custom wake-word detection for AI assistant • Achieved adoption beyond the original ARVSAL use case, recording **584+ downloads** on NPM across diverse voice-assistant

KineSynth-X: Browser-Based Kinematic Synthesis & Simulation Platform *(ME252 Extra-Credit Course Project)* — *Mar'26 - Apr'26*

Objective	• To build a **browser-native platform** for graphical synthesis, simulation, and analysis of planar four-bar mechanisms
Approach	• Implemented a **deterministic state-machine workflow** for multi-step mechanism synthesis using geometric constructions • Designed a decoupled analysis pipeline for computing **velocity, acceleration, and kinematic telemetry** • Developed dynamic visualization modules for **trajectory analysis, workspace exploration, and real-time mechanism**
Result	• Delivered a client-side platform capable of real-time synthesis, simulation, and analysis of **four-bar linkage mechanisms**

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, JavaScript, HTML, CSS
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn
- **Frameworks & Developer Tools:** Electron, Node.js, Git, GitHub, SQLite, Ollama

RELEVANT COURSES

Modern Cryptology *(Ongoing)*	Bayesian Models & Data Analysis	Fundamentals of Computing - I & II
Linear Algebra	Complex Variables	Partial Differential Equations
Economy, Society & Public Policy	Engineering Graphics	Introduction to Electronics

POSITION OF RESPONSIBILITY

Organizer, Fine Arts Cup | Inter-IIT Cultural Meet 8.0 | IIT Kanpur — *Dec'25*

Leadership	• Served as **Organizer** for the Fine Arts Cup vertical during the Inter-IIT Cultural Meet 8.0, overseeing event execution
Execution	• Conducted four Fine Arts Cup competitions: **Live Sketching, Canvas Painting, Charcoal Art, and Costume Design**
Impact	• Facilitated the successful execution of Fine Arts Cup competitions involving participants representing **23 IITs**

Secretary, Fine Arts Club | Media and Cultural Council | IIT Kanpur — *Jul'25 - May'26*

Leadership	• Served as **Secretary of the Fine Arts Club**, coordinating club activities, workshops, performances, and artistic initiatives
Execution	• Organized and coordinated **Fine Arts Club performances** during the Y25 Orientation programme for incoming students • Conducted multiple **fine arts workshops** and community engagement activities to encourage student for creative arts
Impact	• Contributed towards strengthening student participation and engagement in fine arts activities across the institute

EXTRA-CURRICULAR ACTIVITIES

Fine Arts	• Represented **Hall 12** in **Galaxy'26**, securing **1st Position** in both **Live Sketching** and **Comic Art** competitions